

Bifurcation-Guided Replay: Learning at the Success–Failure Boundary of Decision Policies

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Abstract

Policies trained on nominal starts often fail under small deviations from the states encountered during training. Existing curriculum and level-replay methods prioritize difficult or high-regret levels, but they do not explicitly identify where a policy’s behavior transitions from likely success to likely failure. We introduce Bifurcation-Guided Replay (BGR), a general replay method for sequential decision policies. For each replayable decision state, BGR estimates a recovery curve measuring success probability under perturbations of increasing scale, derives a critical radius at which performance crosses a success threshold, and samples training perturbations near this state-conditioned boundary. To make boundary estimation practical, BGR uses active probing, monotone curve fitting, uncertainty-aware refresh, and diversity-preserving replay. We evaluate BGR first in a diagnostic recovery-margin benchmark and then extend the same machinery to procedural decision tasks and robot suffix recovery. BGR improves recovery AUC and shifts critical-radius distributions toward larger margins compared with uniform replay, fixed-radius perturbation training, and failure-focused replay, while preserving clean success.

1 Introduction

A policy can solve a task and still be brittle: a minor shift in object pose, grid layout, browser state, or trajectory prefix may move it outside its recovery basin. Standard success metrics hide this geometry. We ask a different question: for each replayable state, how far can the current policy be perturbed before success becomes failure?

Existing replay and curriculum methods rank examples by difficulty, temporal-difference error, regret, novelty, or failure. These signals are useful, but they do not explicitly model the local response curve around a state. BGR instead estimates a state-conditioned recovery curve and trains near its success–failure transition.

Our contributions are:

- a domain-general replay formalism based on replayable decision states, perturbation families, recovery curves, critical radii, and recovery AUC;

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- an efficient BGR algorithm that actively estimates success–failure boundaries and samples both states and perturbation radii near those boundaries;
- a diagnostic benchmark and experiment pipeline for measuring recovery-margin expansion under controlled replay strategies;
- a path to procedural RL and robot suffix-recovery experiments using the same estimator and priority interfaces.

2 Problem Setting

Let π_θ be a sequential decision policy and let $Y(\tau) \in \{0, 1\}$ denote terminal success for rollout τ . A replayable decision state $x \in \mathcal{X}$ is any object from which evaluation or training can restart, such as an environment seed, simulator checkpoint, task prefix, or robot suffix state.

For each x , a domain-valid perturbation family applies $\tilde{x} = T(x, \delta)$ with $\delta \sim \mathcal{P}_x(\sigma)$, where $\sigma \geq 0$ is a perturbation magnitude. The recovery curve is

$$R_\pi(x, \sigma) = \Pr_{\delta, \tau}[Y(\tau) = 1 \mid \tau \sim \pi(\cdot \mid T(x, \delta))]. \quad (1)$$

For threshold α , the critical radius is

$$r_{\pi, \alpha}(x) = \sup\{\sigma : R_\pi(x, \sigma) \geq \alpha R_\pi(x, 0)\}. \quad (2)$$

We evaluate robustness with recovery AUC,

$$\text{RAUC}_\pi(x) = \frac{1}{\sigma_{\max}} \int_0^{\sigma_{\max}} R_\pi(x, \sigma) d\sigma. \quad (3)$$

3 Bifurcation-Guided Replay

BGR maintains a buffer of replayable states. For each state, it probes perturbation radii, fits a monotone recovery curve, estimates a critical radius, and assigns replay priority to states whose boundaries are feasible, sharp, uncertain, and near a trainable target band.

The priority for a record is a product of gates and soft utilities:

$$P(x) = G(x)U_{\text{boundary}}(x)U_{\text{sharp}}(x)^\lambda U_{\text{unc}}(x)^\beta U_{\text{stale}}(x)^\eta. \quad (4)$$

We mix this distribution with uniform replay to prevent starvation.

Benchmark	Method	Clean	RAUC	Median r_{80}	RAUC AULC
Synthetic margin	BGR	0.8962	0.3756	0.2829	0.3169
Synthetic margin	Uniform	0.8886	0.3659	0.2753	0.2987
Grid margin	BGR	0.9474	0.4350	0.3442	0.3538
Grid margin	Uniform	0.8969	0.3979	0.3322	0.3145
Robot suffix	BGR-Suffix	0.8738	0.4730	0.4534	0.3745
Robot suffix	Uniform suffix	0.8368	0.4854	0.5032	0.3716

Table 1: Current diagnostic results. BGR is strongest in the synthetic and procedural grid-margin settings. In the lightweight robot-suffix simulator, BGR-Suffix improves clean success and sample efficiency over uniform suffix replay, but uniform has higher final object-perturbation RAUC.

Bifurcation-Guided Replay

Initialize replay buffer $\mathcal{B}$ with replayable states.

For each $x \in \mathcal{B}$, probe $\sigma \in \{0, \sigma_{\min}, \sigma_{\text{mid}}, \sigma_{\max}\}$, fit monotone $\hat{R}(x, \sigma)$, and estimate $\hat{r}_\alpha(x)$.

For each training iteration, sample $x \sim q_{\mathcal{B}}(x)$ proportional to BGR priority, sample σ from a mixture centered at $\hat{r}_\alpha(x)$, apply $\delta \sim \mathcal{P}_x(\sigma)$, collect rollout or teacher recovery data, update π_θ , and periodically refresh stale curve estimates.

Figure 1: BGR training loop without domain-specific learner details.

4 Experiments

4.1 Tier-0 Recovery-Margin Benchmark

The initial benchmark creates replayable states with latent feasible radii and policy recovery margins. Success probability follows a sigmoid recovery curve, and training updates near the current margin expand that margin most efficiently. This diagnostic environment tests whether BGR’s estimator and sampler recover the intended behavior before expensive RL or robotics runs.

In the current three-seed diagnostic run, BGR improves final recovery AUC over uniform replay (0.3756 vs. 0.3659), improves the area under the RAUC learning curve (0.3169 vs. 0.2987), and preserves clean success (0.8962 vs. 0.8886). Fixed-radius, failure-only, and PLR-loss proxy baselines underperform in this controlled setting, suggesting that explicit boundary-radius sampling is useful when the benchmark contains learnable success–failure transitions.

4.2 Procedural Grid-Margin Study

We instantiate replayable states in procedural grid environments with resettable seeds and mid-path states. Generated obstacle maps define feasible perturbation neighborhoods through shortest-path reachability, while each replay state has a policy-specific recovery margin. This benchmark isolates whether BGR can expand state-conditioned recovery curves in a non-robotic sequential decision setting.

In a five-seed grid-margin run, BGR outperforms uniform replay on final RAUC (0.4350 vs. 0.3979), RAUC learning-curve area (0.3538 vs. 0.3145), median r_{80} (0.3442 vs. 0.3322), and clean success (0.9474 vs. 0.8969). Failure-only replay, PLR-loss proxy replay, and fixed-radius perturbation training substantially underperform, indicating that neither failure mining nor static perturbation scales recover the same

margin-expansion behavior. A separate tabular grid-policy diagnostic saturated under broad replay and is treated as a negative setup result rather than evidence for the main claim.

4.3 Robot Suffix Study

BGR-Suffix treats a robot demonstration suffix as a replayable decision state. The current lightweight suffix simulator models language-conditioned manipulation families, demonstration suffix phase, object-pose perturbations, end-effector perturbations, teacher feasibility, and clean-demo regularization. Training uses object-pose offsets; end-effector offsets are used as a perturbation-family transfer diagnostic.

In a five-seed suffix run, BGR-Suffix improves clean success over uniform suffix replay (0.8738 vs. 0.8368) and slightly improves RAUC learning-curve area (0.3745 vs. 0.3716), while substantially outperforming clean-only fine-tuning, fixed-radius recovery training, failure-only replay, and loss-priority replay. Uniform suffix replay currently has higher final object-perturbation RAUC (0.4854 vs. 0.4730), higher median r_{80} (0.5032 vs. 0.4534), and slightly higher end-effector transfer RAUC (0.3081 vs. 0.3029). We therefore treat this suffix simulator as a diagnostic BGR-Suffix instantiation rather than final robotics evidence; the next step is a LIBERO/OpenVLA-style simulation run with real task rollouts.

5 Limitations

BGR requires reset or replay access, a meaningful perturbation family, and extra rollouts for boundary estimation. It does not certify robustness, and the term bifurcation is used operationally for an observed success–failure transition rather than a formal dynamical-systems claim.

6 Conclusion

BGR converts recovery-margin measurement into a replay curriculum. By training at feasible success–failure boundaries, it targets the states and perturbation scales where policy improvement should most directly expand local recovery basins.